UNIVERSITY OF KELANIYA – SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science Honours in Management & Information Technology Degree Examination – March/April 2023

Academic Year 2020/2021 – Semester II

Information Technology
INTE 22212 – Software Design Patterns and Frameworks

No. of Questions:4 **No of Pages:3** **Time: 2 1/2 hrs.**

*Answer **ALL** questions. When allocating time for questions, take the marks allocated into consideration. State any assumption that you make.*

*This is an **OPEN BOOK** exam.*

QUESTION 1

(25 Marks)

You are asked to develop a website for travelers to book their foreign trips. Travelers (users) need to book a air ticket, hotel, and taxi. When the user requests an air ticket, a hotel, and a taxi, it should collect data from different web services and send details to the user. The Hotel sub-system will connect to various web services provided by the different hotels (third party). Similarly, the Air tickets sub-service will connect to the airline web services, and the Taxi sub-service will connect to the taxi web services. Please note that third-party organizations have developed these web services; hence web service method signatures can be different from each other.

1.1 Draw a diagram to depict the above scenarios. **[5 Marks]**

1.2 What design pattern can be employed for sub-systems that combine air tickets, hotels, and taxis? Explain the reasons for your selection. **[4 Marks]**

1.2.1 What challenges will you face when you design the system with the selected design pattern? **[3 Marks]**

1.2.2 Explain the mechanism that you will use to overcome those challenges? **[3 Marks]**

1. 3 What design patterns can be employed for connecting the sub-system with different third-party web services? **[4 Marks]**

1.3.1 What challenges will you face when you design the system with the selected design pattern? **[3 Marks]**

1.3.2 Explain the mechanism that you will use to overcome those challenges? **[3 Marks]**

QUESTION 2

(25 Marks)

A Supermarket has different types of pay calculations for different employee categories. The supermarket consists of executive employees, temporary staff, and daily workers. Following are the main salary calculations for each employee category.

Executive Employees: Basic Salary, Bonus & Executive Allowances.

Daily workers: 8 hours per day, and additional hours will be entitled to overtime. More than 200 hours will earn an attendance allowance.

Temporary staff: This category of staff is called when required, and a daily allowance will be paid for the requested dates.

All employees are contributed to Employee Provided Funds by 8%.

2.1 Draw a diagram to depict the above scenario. **[4 Marks]**

2.2 What design pattern(s) can be employed for the above system? Explain the reasons for your selection. **[6 Marks]**

2.3 Draw a class diagram for the above design pattern. **[5 Marks]**

2.4 What challenges will you face when you design the selected design pattern? **[3 Marks]**

2.5 Explain the mechanism that you will use to overcome those challenges? **[3 Marks]**

2.6 “Code duplication can be achieved from the used design pattern in the above scenario.” Do you agree with the above statement? Explain. **[4 Marks]**

QUESTION 3

(25 Marks)

You are hired to design an email spam filter. In the spam filter, multiple sub-spam filters are available depending on the different components of the email, such as email sender, email sent time, email content, and email attachment.

3.1 Sketch a diagram and explain the different components of the email spam filter. **(4 Marks)**

3.2 What design pattern can you follow to design the above system? Explain the reasons. **(5 Marks)**

3.3 How do you prioritize different components of the above email spam filter? Explain your prioritization criteria. **(6 Marks)**

3.4 What would be your suggestion if adding more sub-components to the system is needed? **(5 Marks)**

3.5 How you can extend this design to achieve High Availability (HA) and Disaster Recovery (DR). **(5 Marks)**

QUESTION 4

(5 X 5 = 25 Marks)

Explain the possible design pattern you can suggest for the following scenarios.

4.1 You are chosen to develop a website for an online cake ordering system. Users can select multiple options such as decorations of the cake and colour of the cake. Depending on the user selections, prices and delivery times can differ.

4.2 You are asked to develop a scalable and extensible product-based framework for future needs of the above online cake ordering system.

4.3 You are asked to develop a payroll system. The employee salary calculation should be run only by one user, and all the other connections to the salary systems should be stopped.

4.4 You are developing a learning management system (LMS), and when users read a large content, it is required to load only a couple of pages of the documents rather than the entire document.

4.5 You are developing an LMS that needs different modules to be created depending on the user's needs.